ETFs Simulation with Temporal Memory: A Positive Fractional–Diffusive Volatility Model

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October 2026

Abstract

We develop a conditional-variance model that weights past squared shocks with a finite fractional kernel and pools them across assets through a positive correlation-graph resolvent. We derive its positivity and fixed-parameter moment properties, exact multi-step variance forecasts, and the effective memory of a persistent asymmetric extension. The empirical study covers the S&P 500, eleven sector ETFs and eight country ETFs, with 186 origins per panel during 2023–2026 and horizons of 1, 5 and 21 observations. Persistence, asymmetry and multi-horizon selection improve the initial specification in all six US panel–horizon comparisons. A fixed 50:50 combination with GJR-GARCH lowers pooled QLIKE relative to the refined fractional model and FIGARCH in all nine comparisons. Against GJR alone, it has lower loss in three; all nine paired nominal 95% block-bootstrap intervals include zero at both block lengths. The study therefore establishes neither superiority nor equivalence to GJR. Diffusion has mixed value, and the ranking against weighted Network GARCH reverses between sectors and countries. Evaluation uses retrospective adjusted-price snapshots downloaded in October 2026; chronological row access does not reconstruct historical data availability. The shared temporal kernel and graph operator make the model's impulse response tractable, while the ablations show where these restrictions help or hinder forecasting.

Keywords: fractional memory; conditional variance; graph diffusion; sector ETFs; volatility forecasting; chronological validation.

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1 Introduction

Volatility forecasts can draw on an asset's past shocks and on shocks to related assets. We combine these sources in a conditional-variance model with a fractional temporal filter and diffusion on a correlation graph. The two operators have separate parameters, so we can examine how the memory profile and cross-asset smoothing affect both the model's moments and its forecast errors.

Fractional differencing [9] and FIGARCH [1] motivate the use of slowly decaying temporal weights. Our construction uses a finite, normalized fractional-integral kernel on squared standardized innovations. A correlation graph represents asset relationships, and an implicit diffusion step smooths the resulting variance signals over that graph. The operator has the graph-signal interpretation described by Shuman et al. [18]. A positive intercept and a subunit feedback coefficient complete the discrete variance recursion.

Other parsimonious volatility models distribute information across horizons. The HAR-RV model of Corsi [5] combines daily, weekly and monthly realized-volatility components. Bollerslev et al. [4] show how accounting for variation in measurement error can improve realized-volatility forecasts. These approaches motivate separate attention to memory structure and the quality of the variance proxy. Our data contain daily closes, so HAR-RV and HARQ are related approaches rather than fitted high-frequency benchmarks.

The construction separates three choices: the fractional order allocates weight across lags, diffusion controls how strongly neighboring variance signals are averaged, and feedback controls their overall contribution. An extension adds a persistent state and a larger response to negative innovations. We derive how the state changes the effective memory, so that the effect of persistence can be distinguished from the fractional profile. Sharing the temporal profile and spatial operator across lags reduces the number of parameters relative to unrestricted lag matrices.

1.1 Position relative to network volatility models

Spatial ARCH permits conditional heteroskedasticity to depend on neighboring locations [14], and network log-ARCH combines temporal effects with contemporaneous network interactions [11]. Network GARCH uses graph structure to reduce the dimension of multivariate volatility estimation, with stationarity and QMLE results [21]. Threshold network GARCH also incorporates asymmetric shock responses [15]. More recently, generalized network GARCH includes neighboring shock and variance terms alongside covariance forecasts [22]. The contribution here is the particular restriction to a common positive resolvent at every lag and a normalized fractional temporal profile. Relative to FIGARCH, our finite normalized kernel and subunit feedback define a different recursion. Section 8 includes a fitted FIGARCH reference, and Section 8.4 adds the first-order Network GARCH recursion of Zhou et al. [21], adapted to the available correlation networks. Network log-ARCH, threshold Network GARCH and DCC-GNGARCH remain outside the fitted comparison. The lag-matrix representation below places our model within restricted multivariate ARCH. The proofs apply positive-filter and graph arguments to this parameterization; network dependence and asymmetric responses are established ingredients.

Contributions. We specify a positive fractional–diffusive variance family, establish its fixed-parameter moment properties and derive its effective memory and exact multi-step forecasts. We assess its restrictions through ablations, conventional and network benchmarks, forecast combination and calibration diagnostics. The implementation and reproduction instructions are given in Appendix A. Exact numerical reproduction requires the retained price snapshots.

Roadmap. Sections 2–6 define and evaluate the initial specification; Section 7 adds persistence and asymmetry; Section 8 evaluates country ETFs and additional benchmarks; Section 9 interprets the findings, and the appendices give reproduction details, asset paths and additional diagnostics.

2 Data, notation and prediction target

2.1 Observed prices and returns

Let $P_{t,j} > 0$ denote the closing price of asset j at retained trading observation t, and define

$$r_{t,j} = \log(P_{t,j}/P_{t-1,j}). \tag{1}$$

Write $\mathcal{F}_o = \sigma(P_0, \ldots, P_o)$ for the information generated by observations through origin o in a fixed data vintage. All transformations, graph estimates and parameter choices used to forecast from o depend only on these rows. For the empirical study, this ordering refers to the downloaded snapshot; it does not establish which versions of those rows were available on date o.

The market snapshot was downloaded on 17 October 2026 from Yahoo Finance through `yfinance` 0.2.66, with adjusted prices and an exclusive request end date of 17 October 2026. It contains the index `^GSPC` and the eleven ETFs XLB, XLE, XLF, XLI, XLK, XLP, XLRE, XLU, XLV, XLY and XLC. After inception trimming and alignment, common prices run from 19 June 2018 through 16 October 2026. Prices must be positive, finite, uniquely dated and increasing in date; missing interior observations are not filled.

The index and ETF panel are fitted separately. Adjusted ETF returns and an index price-return series represent different economic quantities. The index is therefore a separate forecasting series, rather than a self-financing replication by the sector funds. Section 9.2 explains how price revisions can affect this retrospective evaluation.

2.2 Risk target and common mean convention

At each parameter refit, estimate a mean μ_j and a positive scale s_j^2 from the observed fitting window. These estimates are shared by all benchmark models until the next refit. Write

$$u_{t,j} = r_{t,j} - \mu_j, \qquad z_{t,j} = u_{t,j}^2/s_j^2. \tag{2}$$

The h-observation evaluation target is

$$Y_{o,h,j} = \sum_{\ell=1}^{h} (r_{o+\ell,j} - \mu_j)^2, \qquad h \in \{1, 5, 21\}. \tag{3}$$

The target uses daily returns. Under the specified constant conditional mean, it is conditionally unbiased for the sum of future conditional variances. A misspecified mean adds squared mean error to that expectation. The empirical task therefore measures forecasts of centered return second moments. Intraday realized variance aggregates high-frequency returns and has its own measurement-error theory [2]; our daily squared-return proxy does not provide that richer measurement.

3 A positive fractional–diffusive model

3.1 Finite fractional-integral memory

For $0 < d \leq 1$, the binomial expansion of $(1 - B)^{-d}$ yields positive coefficients

$$g_0(d) = 1, \qquad g_k(d) = \frac{k - 1 + d}{k}\, g_{k-1}(d) = \frac{\Gamma(k + d)}{\Gamma(d)\Gamma(k + 1)}. \tag{4}$$

With memory length M, normalize these coefficients:

$$p_k(d) = \frac{g_k(d)}{\sum_{v=0}^{M-1} g_v(d)}, \quad m_t = \sum_{k=0}^{M-1} p_k(d) z_{t-1-k}. \tag{5}$$

The weights satisfy $p_k > 0$ and $\sum_k p_k = 1$. For large k, $g_k(d)$ behaves as $k^{d-1}/\Gamma(d)$. Within a fixed window, increasing d spreads weight toward older observations; $d = 1$ gives a uniform moving average. The study fixes $M = 252$. It does not infer d from a Hurst estimator.

The normalized operator is a finite-window analogue of a fractional integral. For a smooth signal q, a physical window T and a refining observation step, the corresponding normalized fractional integral is

$$\mathcal{I}_{d,T} q(t) = \frac{d}{T^d} \int_0^T a^{d-1} q(t - a)\, da. \tag{6}$$

This relation motivates the kernel; the empirical model itself is the finite discrete recursion (5). Finite memory and subunit feedback do not establish asymptotic long-range dependence or convergence to a fractional stochastic PDE.

3.2 Correlation graph and implicit diffusion

Estimate a Ledoit–Wolf covariance matrix [10] after standardizing the fitting-window returns. Convert it to a correlation matrix R and set

$$W_{ij} = \max(R_{ij}, 0)^2 \quad (i \neq j), \qquad W_{ii} = 0. \tag{7}$$

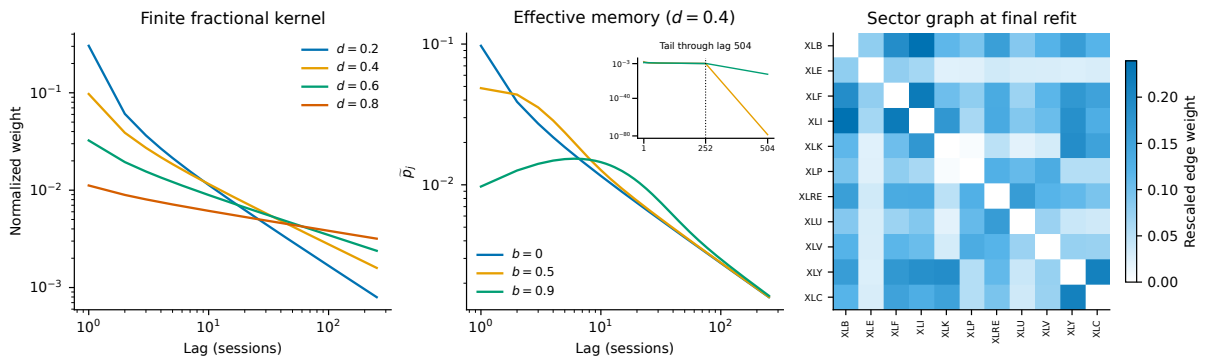


Figure 1: Left: normalized fractional weights over the 252-session window. Center: the effective weights $\widetilde{p}_j$ from (22), illustrated at $d = 0.4$ and $b = 0, 0.5, 0.9$. The main panel shows lags through 252; the inset extends to lag 504. The $b = 0$ curve ends at lag 252; positive persistence adds a geometric tail. These illustrative parameters are not selected fits. Right: the sector graph from the final parameter refit, constructed from rows through that origin.

Let $D = \operatorname{diag}(W\mathbf{1})$ and rescale the graph Laplacian by its mean degree:

$$L = \frac{D - W}{\max(J^{-1} \operatorname{tr} D, 10^{-12})}, \qquad A_\tau = (I + \tau L)^{-1}, \quad \tau \geq 0. \tag{8}$$

For the scalar index, $L = 0$ and $A_\tau = 1$; index results contain no evidence about cross-sector diffusion. For sectors, the update $x = A_\tau m$ is the implicit Euler step $x - m = -\tau L x$ for graph heat diffusion. The diffusion parameter controls smoothing on this estimated network, not a causal contagion rate.

No arbitrary one-dimensional sector ordering is required. Under a permutation matrix Π, the same construction transforms L into $\Pi L \Pi^\top$ and A_τ into $\Pi A_\tau \Pi^\top$. Forecasts inherit this permutation equivariance.

3.3 Conditional variance and return generation

For $0 \leq \eta < 1$, define normalized conditional variance h_t and asset-scale variance v_t by

$$h_t = (1 - \eta)\mathbf{1} + \eta A_\tau m_t, \qquad v_t = s^2 \odot h_t, \tag{9}$$

where $\odot$ denotes elementwise multiplication. Returns obey

$$r_t = \mu + \operatorname{diag}(\sqrt{v_t})\, \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0, \quad \mathbb{E}[\varepsilon_{t,j}^2 \mid \mathcal{F}_{t-1}] = 1. \tag{10}$$

For simulations, the implementation uses temporally independent Gaussian innovation vectors with correlation R. Cross-sectional correlation does not invalidate the marginal moment results below. Gaussian innovations are a simulation assumption, not a claim that market return distributions are Gaussian.

Diffusion acts on standardized variance signals. It preserves their sum, not necessarily the sum of variances in original asset units. This distinction avoids forcing low-volatility and high-volatility sectors to have equal scale. The positive intercept is explicit, and

all simulated paths use the same stated recursion rather than a fitted terminal price correction.

To make the connection with multivariate ARCH explicit, let $S = \mathrm{diag}(s_1^2, \ldots, s_J^2)$. In asset units, the lag coefficient matrices are $C_k = \eta p_k S A_\tau S^{-1}$, giving

$$v_t = (1-\eta)s^2 + \sum_{k=0}^{M-1} C_k\Big(u_{t-1-k} \odot u_{t-1-k}\Big).$$

All C_k are nonnegative, and $\rho(\sum_k C_k) = \eta$. Fractional weighting and graph diffusion constrain these matrices to an interpretable shared structure.

4 Mathematical properties

Proposition 1 (Positive, nonamplifying graph diffusion)**.** *For a symmetric nonnegative adjacency matrix and $\tau \geq 0$, A_τ is entrywise nonnegative and doubly stochastic. Its eigenvalues lie in $(0, 1]$. For any real vector x,*

$$\min_j x_j \leq (A_\tau x)_i \leq \max_j x_j, \qquad (A_\tau x)^\mathsf{T} L (A_\tau x) \leq x^\mathsf{T} L x. \tag{11}$$

Proof. The Laplacian is symmetric positive semidefinite and $L\mathbf{1} = 0$. Hence $I + \tau L$ is invertible and $A_\tau \mathbf{1} = \mathbf{1}$. For $b \geq 0$, a negative minimum of the solution to $x + \tau L x = b$ would make the corresponding left-hand component negative, a contradiction. Hence $A_\tau \geq 0$. Symmetry gives unit column sums. The componentwise bound follows from convex averaging. Diagonalizing L gives eigenvalues $(1 + \tau\lambda_i)^{-1}$ for A_τ; the energy inequality follows from $\lambda_i/(1 + \tau\lambda_i)^2 \leq \lambda_i$. □

Proposition 2 (Positivity and fixed-parameter moment stability)**.** *Fix $M < \infty$, $s_j^2 > 0$ and $0 \leq \eta < 1$. For nonnegative observed squared innovations, $v_{t,j} \geq (1-\eta)s_j^2$. If the innovation vectors are iid over time with zero means and unit marginal second moments, the normalized squared-return recursion admits a unique causal stationary solution with finite first moments, and its stationary marginal expectations satisfy $\mathbb{E}z_t = \mathbf{1}$. It follows that the corresponding stationary returns have variance s_j^2 and finite second moments $\mu_j^2 + s_j^2$.*

Proof. The lower bound follows from nonnegative weights and nonnegative diffusion. If initial component expectations are bounded by $C \geq 1$, then $\mathbb{E}h_{t,j} \leq (1-\eta) + \eta C \leq C$, and $\mathbb{E}z_{t,j} = \mathbb{E}h_{t,j}$. For two histories driven by the same innovation sequence, put $e_t = \mathbb{E}\|z_t - z_t'\|_1$. Independence of the current innovations from the past, unit marginal second moments and the unit column sums of A_τ give

$$e_t \leq \eta \sum_{k=0}^{M-1} p_k e_{t-1-k}. \tag{12}$$

The largest expected discrepancy contracts by at least η each time an entire block of M initial lags is replaced. Backward iteration from a bounded deterministic history is therefore Cauchy in L^1, yielding a causal stationary limit; the same bound gives uniqueness among finite-first-moment solutions. In stationarity, $b = \mathbb{E}z_t$ solves $b = (1-\eta)\mathbf{1} + \eta A_\tau b$. Since $\rho(\eta A_\tau) < 1$, its unique solution is $b = \mathbf{1}$. □

The moment result concerns the fixed-parameter recursion, rather than the sequence produced by rolling estimation. It supplies finite return second moments; fourth moments and a limit distribution for empirical losses would require additional conditions. Finite-window contraction also distinguishes the fractional shape of the weights from asymptotic long-range dependence.

5 Forecasting and estimation

5.1 Exact multi-step marginal moments

At origin o, let $\bar{z}_{o+\ell|o}$ equal observed z for nonfuture lags, and recursively replace future squared innovations by their conditional expectations:

$$\bar{h}_{o+\ell|o} = (1-\eta)\mathbf{1} + \eta A_\tau \sum_{k=0}^{M-1} p_k \bar{z}_{o+\ell-1-k|o}, \qquad \bar{z}_{o+\ell|o} = \bar{h}_{o+\ell|o}. \tag{13}$$

Linearity and unit innovation second moments make this recursion exact under the fitted variance model. The aggregate forecast is

$$\widehat{V}_{o,h} = s^2 \odot \sum_{\ell=1}^{h} \bar{h}_{o+\ell|o}. \tag{14}$$

With the model's conditional mean, centered increments are martingale differences, so (14) also gives the marginal conditional variance of cumulative future log-returns. Future variances remain random, however, and their sum does not give a Gaussian predictive law for returns. Our forecasts target variance directly.

5.2 Chronological selection

Forecast origins occur every five retained observations on a grid beginning at price position 252; scoring begins in 2023. At the first scored origin and whenever at least 21 observations have elapsed since the last refit, parameters are updated. This gives a 25-observation refit spacing on the five-observation grid, with 38 refits per panel in the reported experiment.

At a refit origin, reserve the final 252 observed returns for validation. Means, scales and graph structure used to compute validation predictions are estimated strictly before that block, using at most 1,008 returns. As validation advances, already observed squared returns may update the memory state. Their future counterparts are never accessed. Select the minimum pooled one-step QLIKE over

$$d \in \{0.2, 0.4, 0.6, 0.8\}, \qquad \eta \in \{0.8, 0.95, 0.99\}, \tag{15}$$

$$\tau \in \{0, 0.25, 1, 4\}. \tag{16}$$

The index fixes $\tau = 0$. Stable candidate order resolves exact ties. After selection, refit means, scales and the graph on the last 1,008 available returns, including validation observations, and forecast subsequent targets. The historical period was available during earlier research iterations, so this chronological evaluation remains a development study.

The selected parameters serve all three horizons, although selection minimizes only one-step loss. That mismatch motivates the multi-horizon refinement in Section 7. All methods are scored at the same origins, excluding end blocks without complete targets for every horizon.

5.3 Benchmarks and ablations

1. **Historical variance:** the current fitting-window sample variance, held constant over each forecast horizon.
2. **EWMA:** squared-innovation averaging with decay selected from $\{0.9, 0.94, 0.97, 0.99\}$ on the same validation block. Its future conditional variance remains constant under its own update equation.
3. **GARCH(1,1):** independent asset models estimated by Gaussian quasi-maximum likelihood [3]. Positivity is imposed and $\alpha + \beta < 0.999$. Two fixed initializations are used; all 456 selected asset-level fits report convergence.
4. **Fractional only:** the proposed kernel and feedback with $\tau = 0$, with its own validation-selected d and η.
5. **Exponential plus graph:** replace the fractional average by EWMA while retaining positive feedback and graph diffusion. Its decay, feedback and diffusion grids have the same candidate count as the fractional graph model.

The last two comparisons isolate graph smoothing and the shape of the memory kernel. Selecting each variant on its own validation losses allows its remaining parameters to adapt when a component is removed.

6 Evaluation of the initial specification

6.1 Loss and dependence

The primary score is a normalized QLIKE loss,

$$\mathcal{L}_{o,h,j} = \log\left(\frac{\widehat{V}_{o,h,j}}{h s_j^2}\right) + \frac{Y_{o,h,j}}{\widehat{V}_{o,h,j}}. \tag{17}$$

The normalization is known at the origin and common to all methods. It changes the level, not paired comparisons. QLIKE is appropriate for conditional second-moment forecasting with noisy, conditionally unbiased proxies [16, 17]. Scores can be negative, and ratios of score levels are not reported as percentage improvements. A normalized squared-error diagnostic is also retained in the machine-readable results.

There are 186 common origins from 9 January 2023 to 17 September 2026, with targets through 16 October 2026. The sector panel contributes 2,046 asset–origin pairs per horizon, which must not be treated as independent replications. Twenty-one-observation targets overlap. Paired circular block-bootstrap ranges use 2,000 replicates, seed 1729,

	Index			Sectors		
Horizon	1	5	21	1	5	21
Historical	0.8368	0.6695	0.6737	0.8167	0.7187	0.7222
EWMA	0.8397	0.4576	0.5602	0.7195	0.5914	0.6251
GARCH	**0.6004**	**0.3640**	**0.4714**	**0.6297**	**0.5499**	0.6021
Fractional only	0.6514	0.4223	0.5467	0.6549	0.5638	0.6175
Exponential + graph	0.6690	0.3767	0.4841	0.6524	0.5542	**0.5953**
Fractional + graph	0.6514	0.4223	0.5467	0.6405	0.5627	0.6196

Panel	Horizon	Fractional − GARCH	Nominal 95% interval
Index	1	+0.05103	[−0.07687, +0.17028]
Index	5	+0.05823	[−0.00257, +0.12630]
Index	21	+0.07532	[−0.01504, +0.20011]
Sectors	1	+0.01084	[−0.03822, +0.04668]
Sectors	5	+0.01287	[−0.00534, +0.03247]
Sectors	21	+0.01752	[−0.01941, +0.06475]

Table 1: Mean normalized QLIKE at 1, 5 and 21 observations. Lower is better. All methods use identical origins and the same mean convention. The index has no cross-sector diffusion, so its two fractional rows coincide. Bold marks the lowest unrounded loss among the displayed models in each column, including ties. The lower panel adds fractional-plus-graph minus GARCH differences and nominal 95% intervals using ten-origin blocks.

and blocks of five or ten origins. Entire sector vectors are resampled together. Each replicate averages the paired loss difference, with negative values favoring the proposed forecast. The 2.5th and 97.5th percentiles give the reported nominal 95% intervals. This resampling preserves dependence within each block and across assets; it does not adjust for model selection, multiple comparisons, nonstationarity or previous researcher exposure. An interval containing zero leaves the direction unresolved under this procedure; it is not an equivalence test. Appendix E adds paired predictive-accuracy tests following Diebold and Mariano [6] and calibration regressions, with dependence-aware covariance estimates. The statistical interpretation in this paragraph applies to all subsequent loss comparisons.

6.2 Pooled comparison

Table 1 shows lower mean loss for fractional diffusion than tuned EWMA at every panel–horizon combination. The one-session differences are −0.18834 for the index and −0.07899 for sectors. GARCH has lower mean QLIKE in all six cases. For sectors, the proposed method's excess losses relative to GARCH are 0.01084, 0.01287 and 0.01752 across the three horizons.

Graph diffusion changes the one-session sector score from 0.6549 to 0.6405. Its gain is much smaller at five observations, and the no-diffusion model is slightly better at 21 observations. Nonzero diffusion is selected at 31 of 38 sector refits, but validation selection

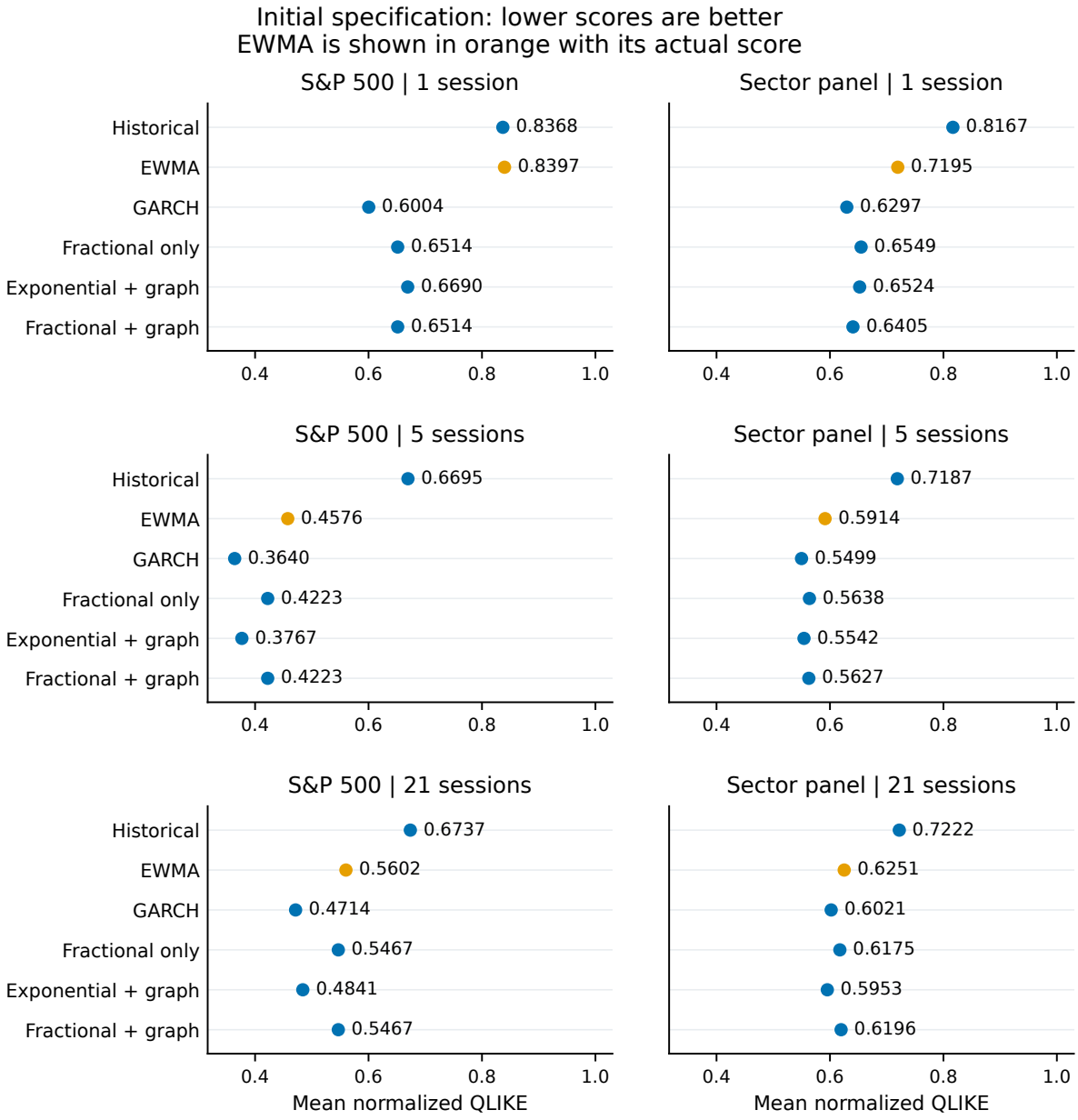


Figure 2: Mean normalized QLIKE for the initial specification and references. Lower values indicate smaller loss. Each point and its numerical label show the actual score, including EWMA in orange. The six panels share the same horizontal scale and reproduce Table 1; all methods use identical origins within each panel and horizon.

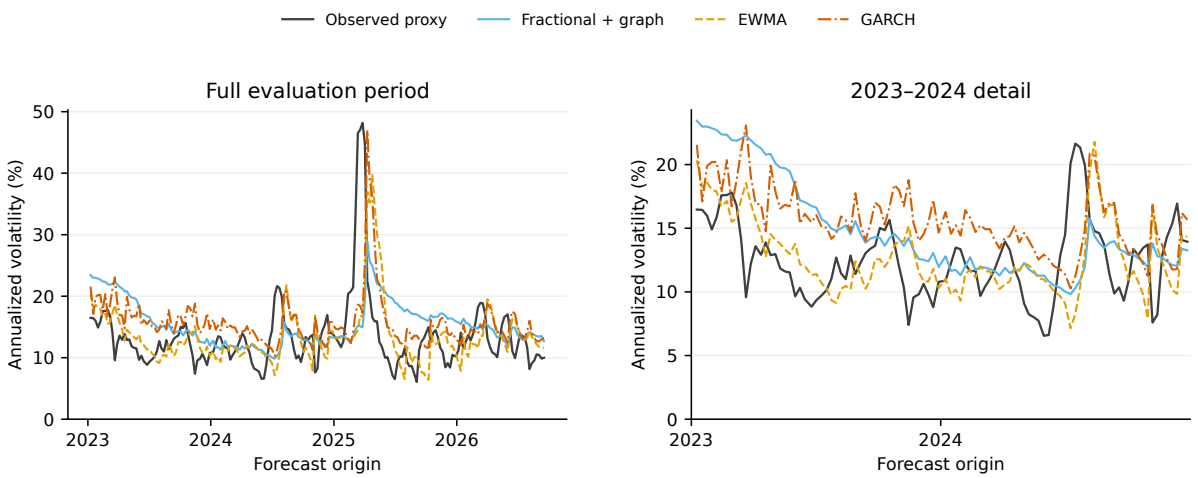


Figure 3: Index risk forecasts over 21 subsequent observations, displayed in annualized volatility units. The black series is the subsequently observed squared-return proxy, plotted at its forecast origin for comparison; it was not available then. The left panel retains the full period; the right enlarges 2023–2024 with its own vertical range. Both use the same saved observations and forecasts.

is not itself evidence of outer-sample predictive benefit. The exponential graph model outperforms fractional diffusion at five and 21 observations in both panels; the fractional kernel has the lower one-session mean in both panels. Memory shape matters, but no shape dominates all horizons.

With ten-origin blocks, the nominal 95% range for the one-session sector difference from EWMA is $[-0.26077, 0.03470]$. Relative to no diffusion, it is $[-0.03672, 0.00145]$. Both include zero. The difference from GARCH is 0.01084, with range $[-0.03822, 0.04668]$. These intervals leave all three rankings unresolved. Full intervals for both block lengths accompany the tables.

6.3 Variation over time

Average improvements are not uniform over calendar years. For the index, the proposed method loses to EWMA at all horizons in 2023, wins at all horizons in 2024 and 2026, and has mixed results in 2025. For sectors, it loses at all horizons in 2023 and improves on EWMA in the later years. The 2025 index one-session difference of -0.5367 contributes substantially to the pooled advantage over EWMA.

Figure 3 shows the corresponding forecast paths. Asset-level losses are retained in the local origin records, and aggregate annual scores are included in the manuscript bundle.

7 Persistence and asymmetric innovations

7.1 Specification and moment properties

The initial model responds identically to positive and negative innovations and selects parameters using only one-step losses. We retain its fractional kernel and graph diffusion while allowing a persistent variance state and asymmetric shock contributions. Define

$$x_{t,j} = z_{t,j}\frac{1+\delta\mathbf{1}\{u_{t,j}<0\}}{1+\delta/2}, \tag{18}$$

$$h_t = (1-b)(1-\eta)\mathbf{1} + bh_{t-1} + (1-b)\eta A_\tau \sum_{k=0}^{M-1} p_k x_{t-1-k}, \tag{19}$$

with $v_t = s^2 \odot h_t$, $0 \le b < 1$ and $\delta \ge 0$. The parameter b smooths the variance response. With $\delta = 1$, a negative innovation has twice the contribution of a positive innovation of the same magnitude.

For the asymmetric recursion, assume conditionally symmetric innovations with unit second moments. Conditional symmetry strengthens the moment assumptions: symmetry gives $\mathbb{E}[\varepsilon_{t,j}^2\mathbf{1}\{\varepsilon_{t,j}<0\}] = 1/2$ and therefore $\mathbb{E}[x_t \mid \mathcal{F}_{t-1}] = h_t$. Gaussian simulation innovations satisfy the assumption; it is not asserted as an empirical property of market shocks. Future expected variances are obtained by the same substitution as in Section 5.1, retaining the additional state h_{t-1}.

Unit initial variance and unit presample transformed squares give the floor $v_{t,j} \ge (1-\eta)s_j^2$ by induction. For fixed parameters and iid symmetric innovations, the shared-noise argument for Proposition 2 now gives, for discrepancies between normalized variance processes,

$$e_t \le be_{t-1} + (1-b)\eta \sum_{k=0}^{M-1} p_k e_{t-1-k}, \qquad b + (1-b)\eta < 1. \tag{20}$$

The maximum expected discrepancy over the finite lag state contracts over successive M-observation blocks. The same backward-iteration argument yields a unique causal finite-first-moment stationary variance process. Its mean is $\mathbf{1}$, so return variance is s_j^2. The recursive state adds exponentially decaying persistence; it does not establish asymptotic long-range dependence.

7.2 Effective memory and graph modes

The persistence state has an exact filter interpretation. For fixed parameters, unroll (19) after removing its initialization transient to obtain

$$h_t = (1-\eta)\mathbf{1} + \eta A_\tau \sum_{j=1}^{\infty} \widetilde{p}_j x_{t-j}, \tag{21}$$

$$\widetilde{p}_j = (1-b)\sum_{k=0}^{\min(M-1,j-1)} p_k b^{j-1-k}. \tag{22}$$

Indeed, repeated substitution expands the state into a geometric sum; grouping terms with the same innovation lag yields (22). Since all terms are nonnegative, summing in

either order gives $\sum_{j\geq 1}\widetilde{p}_j = (1-b)\sum_k p_k \sum_{\ell\geq 0} b^\ell = 1$. For $0 < b < 1$ and $j \geq M$, $\widetilde{p}_{j+1} = b\widetilde{p}_j$. At $b = 0$ the weights reduce to p_{j-1} for $1 \leq j \leq M$, with zero tail. The refinement geometrically smooths the finite fractional kernel; its infinite tail, when present, is exponential rather than hyperbolic.

With $L = U\Lambda U^\mathsf{T}$, the graph-mode coefficient at lag j is $\eta\widetilde{p}_j/(1+\tau\lambda_\ell)$. Temporal weighting and spatial smoothing are separable: every lag shares the same graph attenuation. This clarifies both interpretability and a restriction relative to more general network lag matrices. A numerical impulse-response test verifies the identity against the implemented filtering recursion.

7.3 Recorded follow-up protocol

The refinement was designed after inspecting the initial study. `paper/refinement_protocol.json` records its rules before candidate evaluation on the reused historical targets. At each refit, use the same 252-observation validation block and fit nuisance parameters strictly before it. Retain only validation origins whose complete 21-observation targets finish by the refit date. Minimize the equal-weight mean of normalized QLIKE at horizons 1, 5 and 21, using all eligible daily origins. A single selected specification serves all outer horizons.

Retain the earlier grids for d, η and τ, and add $b \in \{0, 0.5, 0.9\}$ and $\delta \in \{0, 1\}$. There are 288 candidates for sectors and 72 for the scalar index per kernel family. A symmetric variant fixes $\delta = 0$; a graph ablation fixes $\tau = 0$; a kernel ablation uses normalized exponential weights truncated to the same 252 observations, with the earlier four decay values. Each ablation has its own validation selection. The larger candidate set increases selection risk despite chronological ordering.

An additional GJR-GARCH reference permits asymmetric shock responses [7]. Its normalized variance satisfies $h_t = \omega + (\alpha + \gamma\mathbf{1}\{u_{t-1} < 0\})z_{t-1} + \beta h_{t-1}$. Independent asset fits use Gaussian QMLE, two fixed starts, nonnegative coefficients and $\alpha + \beta + \gamma/2 < 0.999$. All 456 selected fits report convergence. Multi-step moments also require conditional symmetry. Earlier reference forecasts are recomputed; all 40,176 original forecast records and loss values match exactly, so comparisons use unchanged targets and scales.

7.4 Follow-up results

Table 2 shows lower pooled loss for persistent asymmetric fractional diffusion than the initial model in all six comparisons. At horizons 1, 5 and 21, respectively, the pooled scores change as follows:

$$\begin{aligned} \text{Index}: &\quad (0.6514, 0.4223, 0.5467) \longrightarrow (0.5819, 0.3574, 0.4977), \\ \text{Sectors}: &\quad (0.6405, 0.5627, 0.6196) \longrightarrow (0.6242, 0.5478, 0.6043). \end{aligned}$$

The refinement also has lower loss than GARCH at one and five observations in both panels. GARCH remains better at 21 observations, while GJR-GARCH is better at every panel–horizon combination. Persistence, asymmetry and multi-horizon selection improve the fractional family jointly; the comparison with GJR shows the remaining gap to a conventional asymmetric specification.

	Index			Sectors		
Horizon	1	5	21	1	5	21
Fractional + graph	0.6514	0.4223	0.5467	0.6405	0.5627	0.6196
Persistent symmetric	0.6143	0.3815	0.5162	0.6447	0.5582	0.6116
Refined fractional	0.5819	0.3574	0.4977	0.6242	0.5478	0.6043
Asymmetric, no graph	0.5819	0.3574	0.4977	0.6317	0.5474	0.6046
Asymmetric exponential	0.6223	0.3516	0.4785	0.6656	0.5588	0.5980
GARCH	0.6004	0.3640	0.4714	0.6297	0.5499	0.6021
GJR-GARCH	**0.5251**	**0.3128**	**0.4360**	**0.6053**	**0.5401**	**0.5931**

Table 2: Mean normalized QLIKE for the refinement, its ablations and references. "Fractional + graph" denotes the initial variance specification. All rows use the same 186 origins. Lower is better; bold marks each column's minimum.

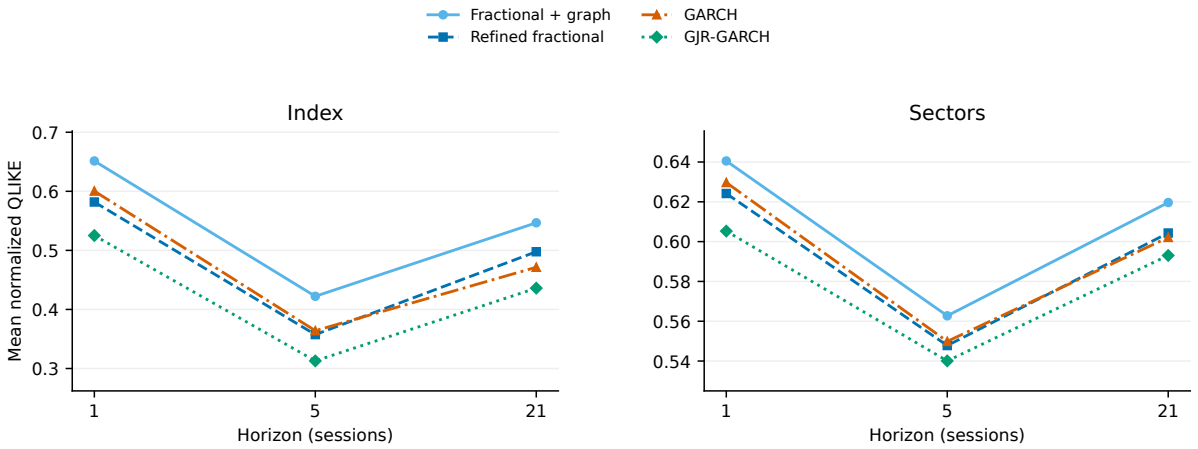


Figure 4: Pooled losses before and after the refinement, alongside GARCH and GJR-GARCH. The asymmetric reference remains stronger in every displayed case.

Using ten-origin blocks, nominal 95% ranges for the change relative to the initial fractional model exclude zero in five of six comparisons; the exception is one-session sector forecasting. For the index, the one-session change is -0.06952 with range $[-0.14531, -0.00876]$. All ranges comparing the refinement with GARCH include zero. Their interpretation follows the exploratory bootstrap convention in Section 6.1.

Asymmetry is selected at all 38 index refits and 35 sector refits; nonzero variance persistence is selected at every refit. The graph is selected with nonzero strength at 18 sector refits. Relative to its graph ablation, the refinement improves the one-session sector mean by 0.00752, slightly worsens the five-session mean and changes the 21-session mean little. The exponential ablation is stronger for index horizons 5 and 21 and sector horizon 21. The ranking depends on both the component and the horizon. Annual improvements also vary: relative to the first specification, index loss rises at 21 sessions in 2024, and sector loss rises at 5/21 sessions in 2024 and at one session in 2025. All annual results remain in the public tables.

8 Extension to country ETFs and additional benchmarks

8.1 A specification fixed before the new download

The previous two studies reuse US history inspected during development. To assess transfer to another universe, a third protocol fixed eight country ETFs before downloading their prices: EWA (Australia), EWC (Canada), EWG (Germany), EWH (Hong Kong), EWJ (Japan), EWL (Switzerland), EWQ (France) and EWU (United Kingdom). The long-established US-listed iShares equity funds were selected for geographical spread. The request covers 1 January 2018 through 16 October 2026, using the same adjusted-price convention. All eight assets are required; missing prices are not filled and assets are not removed after seeing results.

The unchanged fractional implementation, parameter grids, validation windows, refit schedule and three horizons are frozen in `paper/external_protocol.json`, with source hashes. No external loss is used to revise that specification. The timestamp is a local audit record. The country panel had not been evaluated in this project, but it shares global shocks, US trading hours and dollar denomination with the development assets. This experiment tests transfer to another asset universe on the same historical period; it does not supply independent or prospective confirmation.

The protocol amendment records an array-layout correction required by the FIGARCH implementation. Strided panel columns are copied to C-contiguous arrays before forecasting. This correction preceded the FIGARCH comparison results and left the model and evaluation rules unchanged; both the original protocol and amendment are retained.

8.2 Additional forecaster and fractional benchmark

We also evaluate a fixed combination of variance forecasts:

$$\widehat{V}^{\mathrm{ens}}_{o,h,j} = \tfrac{1}{2}\widehat{V}^{\mathrm{frac}}_{o,h,j} + \tfrac{1}{2}\widehat{V}^{\mathrm{GJR}}_{o,h,j}\,. \tag{23}$$

The fractional component is the persistent asymmetric model with its selected graph diffusion. The one-half coefficient preserves positivity and is retained across assets, origins and horizons. This combination acts on the forecasts; the standalone fractional recurrence remains unchanged. We compute QLIKE from the combined variance, rather than averaging component losses.

Simple forecast combinations can limit weight-estimation error, a central theme in the review by Timmermann [19]. Equal weighting gives each component the same contribution without estimating an additional parameter from this relatively short evaluation history. The weight was specified before the country-ETF download. Learning a weight would require a further chronological validation layer and would introduce estimation uncertainty. Since GJR is already a strong component, the combination must also be compared with GJR alone to assess whether the fractional forecasts add useful information. Both components are reported alongside their combination.

FIGARCH(1,d,1) [1] is added as an estimated benchmark using `arch` 7.2.0, Gaussian QMLE and the library's 1,000-lag ARCH approximation. Each asset is centered and standardized using the same origin-fitted mean and scale as the other methods. Fit on the preceding 1,008 returns at the same refit dates, with default starting estimates and SLSQP limited to 1,000 iterations. Between refits, keep parameters and the fitting backcast fixed while updating on observed returns. Analytic forecasts are transformed back to asset units and summed over the same horizons. Nonfinite results abort the experiment; finite nonconverged fits are reported rather than silently replaced. FIGARCH's approximation and initialization differ from our normalized 252-lag kernel and are recorded explicitly.

The same ensemble and FIGARCH reference are appended to the recorded US forecasts after checking their code and input hashes. Those comparisons remain exploratory. Every table uses identical targets within its panel; pooled, annual and asset-level losses are retained. Paired uncertainty ranges use the same whole-panel circular bootstrap and the same nominal interpretation as the earlier study.

8.3 Results

The international panel contains 2,210 prices per asset. Evaluation has 186 origins from 6 January 2023 to 16 September 2026, with complete targets through 15 October 2026 and 38 parameter refits. Its five-observation grid has a different calendar phase from the US grid; comparisons are paired within panels. All 760 FIGARCH asset fits across the three panels converge, as do all 304 international GARCH and 304 GJR-GARCH fits.

The unchanged fractional model obtains international QLIKE $(0.52898, 0.66359, 0.67147)$ at horizons $(1, 5, 21)$, below both EWMA and FIGARCH at every horizon. However, GARCH and GJR-GARCH are stronger at all three. The no-graph ablation has lower loss throughout, while the exponential kernel is better at horizons 5 and 21. Thus the

	Countries		
Horizon	1	5	21
EWMA	0.5321	0.6700	0.6823
GARCH	0.5287	0.6587	0.6672
GJR-GARCH	**0.5083**	**0.6496**	0.6623
FIGARCH	0.5438	0.6671	0.6730
Asymmetric, no graph	0.5234	0.6633	0.6701
Asymmetric exponential	0.5290	0.6526	0.6653
Refined fractional	0.5290	0.6636	0.6715
Fixed 50:50 ensemble	0.5131	0.6509	**0.6613**

Table 3: Mean normalized QLIKE on the eight pre-specified country ETFs. The fractional row is the unchanged persistent asymmetric model. The fixed ensemble averages its variance forecasts with GJR-GARCH. Lower is better; bold marks the minimum in each column.

second universe does not support a general benefit from diffusion or a uniform fractional-kernel advantage.

The fixed ensemble lowers those fractional scores to $(0.51307, 0.65094, 0.66134)$, improving all three external pooled comparisons. Its asset-level loss is lower in 22 of 24 country–horizon comparisons, but annual pooled loss worsens relative to the standalone fractional model at horizons 5 and 21 in 2024. Across the US and international panels, the ensemble improves on both the standalone fractional model and FIGARCH in all nine panel–horizon comparisons. These are loss differences, not percentages of an arbitrarily normalized score.

Relative to GJR-GARCH, the ensemble wins only for US sectors at 5 and 21 observations and for the international panel at 21 observations. The international 21-session difference is -0.00092, with ten-origin-block nominal 95% range $[-0.01174, 0.01110]$. All nine ensemble-versus-GJR ranges include zero. For international one-session forecasts, the change from the standalone fractional model is -0.01590; the interval is $[-0.03100, -0.00219]$.

Compared with FIGARCH, the difference is -0.03073. Its interval runs from -0.05536 to -0.01145. These descriptive intervals support reporting the observed improvement without claiming reliable superiority over GJR-GARCH. Full annual, asset-level and paired comparisons remain in the public outputs.

8.4 Direct comparison with Network GARCH

A follow-up benchmark evaluates the first-order recursion of Zhou et al. [21], equation (2.2), on the same sector and country panels:

$$h_t^N = \omega \mathbf{1} + \alpha z_{t-1} + \lambda B z_{t-1} + \beta h_{t-1}^N, \qquad \omega > 0, \quad \alpha, \lambda, \beta \geq 0, \quad \alpha + \lambda + \beta < 0.999. \tag{24}$$

Here B row-normalizes the positive squared correlation weights already used to construct the fractional model's graph. Its diagonal and isolated rows are zero. A sensitivity benchmark replaces these weights by binary edges at $R_{ij} \geq 0.5$ before normalization; a pooled GARCH control sets $\lambda = 0$. The coefficients are shared across the standardized

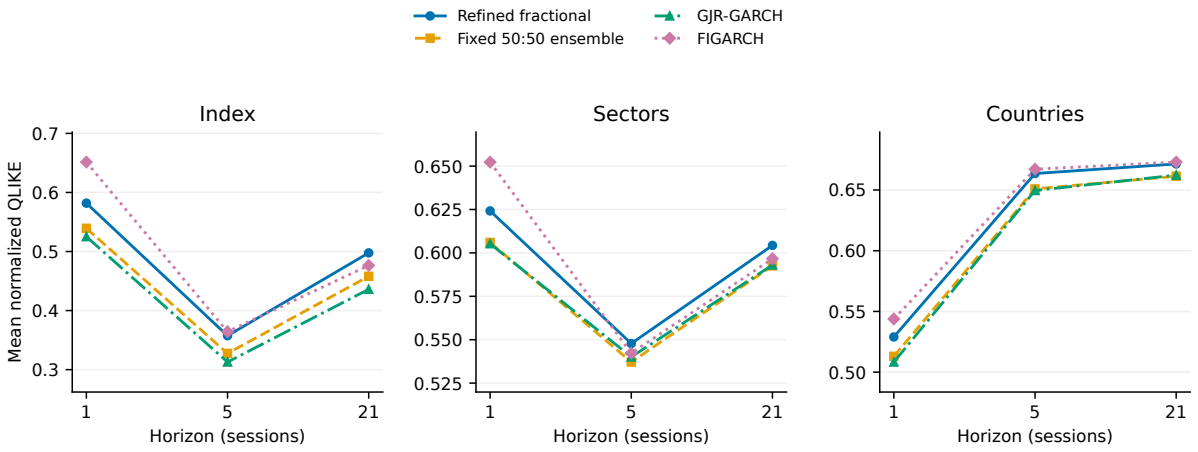


Figure 5: Standalone fractional forecasts, the fixed ensemble, GJR-GARCH and FIGARCH. The index and sector panels reuse the development snapshot; the international panel follows the separately recorded external-universe protocol.

assets. These choices adapt the published recursion to estimated correlation networks rather than the original shareholder network. They do not reproduce the original paper's empirical application or its assumptions of an exogenous network.

At each of the 38 retained refits per panel, marginal Gaussian QMLE uses the preceding 1,008 returns, the common fitted means and scales, and initial normalized variance one. Three fixed starting points are optimized with analytic gradients; the fit with the lowest finite training objective is retained. Observed returns update the state between refits. The first future variance follows (24); subsequent conditional expectations replace future z_t by its conditional mean h_t^N. Daily forecasts are summed over the original horizons and evaluated on identical targets. All 228 selected fits and all 684 optimization starts converge.

The new protocol was recorded after examining the earlier results and before fitting these benchmarks. It is an exploratory comparison on reused data; the fractional forecasts, combination weights and previous studies are unchanged. The index is excluded because a single asset supplies no network comparison. Annual scores, asset-level scores and all fitting attempts are retained with the protocol.

The weighted Network GARCH benchmark has lower pooled loss than the standalone fractional model at all three sector horizons; the ordering reverses at all three country horizons. Every paired 95% range for these six differences includes zero at both block lengths. The fixed ensemble has lower pooled loss than either network variant and the pooled control in all six comparisons. Its one-session country difference against weighted Network GARCH is -0.02660, with ten-origin-block range $[-0.04645, -0.00747]$; the five-session range excludes zero with ten-origin blocks but includes it with five-origin blocks. The other four ranges include zero at both lengths.

Relative to pooled GARCH, weighted network feedback helps the sector mean at one and five sessions, but worsens it at 21 sessions and at all country horizons. All six paired ranges include zero. Binary network losses are higher than weighted network losses in each comparison. This additional experiment therefore supports a context-dependent role for

	Sectors			Countries		
Horizon	1	5	21	1	5	21
Refined fractional	0.6242	0.5478	0.6043	0.5290	0.6636	0.6715
Fixed 50:50 ensemble	0.6060	**0.5371**	**0.5925**	0.5131	0.6509	**0.6613**
GJR-GARCH	**0.6053**	0.5401	0.5931	**0.5083**	**0.6496**	0.6623
FIGARCH	0.6523	0.5422	0.5966	0.5438	0.6671	0.6730
Network GARCH (weighted)	0.6149	0.5465	0.6004	0.5397	0.6642	0.6741
Network GARCH (binary)	0.6172	0.5490	0.6029	0.5401	0.6648	0.6750
Pooled GARCH	0.6321	0.5513	0.5997	0.5296	0.6614	0.6725

Table 4: Mean normalized QLIKE for the separately recorded network comparison. The columns retain the same 1-, 5- and 21-session targets. Network GARCH uses shared coefficients; the pooled control removes its network term. Lower is better; bold marks the minimum in each column.

cross-asset feedback while leaving GJR-GARCH a strong reference.

8.5 Observed evolution and rolling predictions

Aggregate losses summarize accuracy but do not show how forecasts respond over time. Figures 6 and 7 compare the recorded predictions with the subsequently observed squared-return targets over the full evaluation period. For readability, both are expressed in annualized volatility units:

$$\widehat{\sigma}^{\text{ann}}_{o,h,j} = 100\sqrt{\frac{252}{h}\widehat{V}_{o,h,j}}, \qquad \sigma^{\text{proxy}}_{o,h,j} = 100\sqrt{\frac{252}{h}Y_{o,h,j}}. \tag{25}$$

The display transforms variance forecasts and noisy targets to volatility units. The square root of expected variance generally differs from expected volatility. At one session, the observed series is a scaled absolute centered return and is especially noisy.

The time-series panels use the *end of the target window* on the horizontal axis. A prediction plotted at $o + h$ is computed from rows through o in the recorded snapshot. Curves join the retained origins, spaced five observed sessions apart; they are not a new forecast issued every day. No further fitting, time smoothing, clipping of peaks or favorable subperiod selection is applied. The 21-session target windows overlap.

For the multi-asset displays, average $\widehat{V}_{o,h,j}$ and $Y_{o,h,j}$ over all assets before applying (25). Averaging volatilities directly would give a different quantity. The panel curves can also mask opposing asset-level errors. Appendix D therefore retains the complete set of 21-session paths for every sector and country ETF. These visual diagnostics complement the common-target losses and paired comparisons; visual tracking alone does not establish predictive superiority.

The index curves also reveal limitations hidden by pooled scores. The abrupt 2025 increase is incompletely anticipated, and the persistent forecasts can remain elevated after the observed proxy falls. Smoother predictions should not be interpreted as pointwise reproduction of the realized fluctuations.

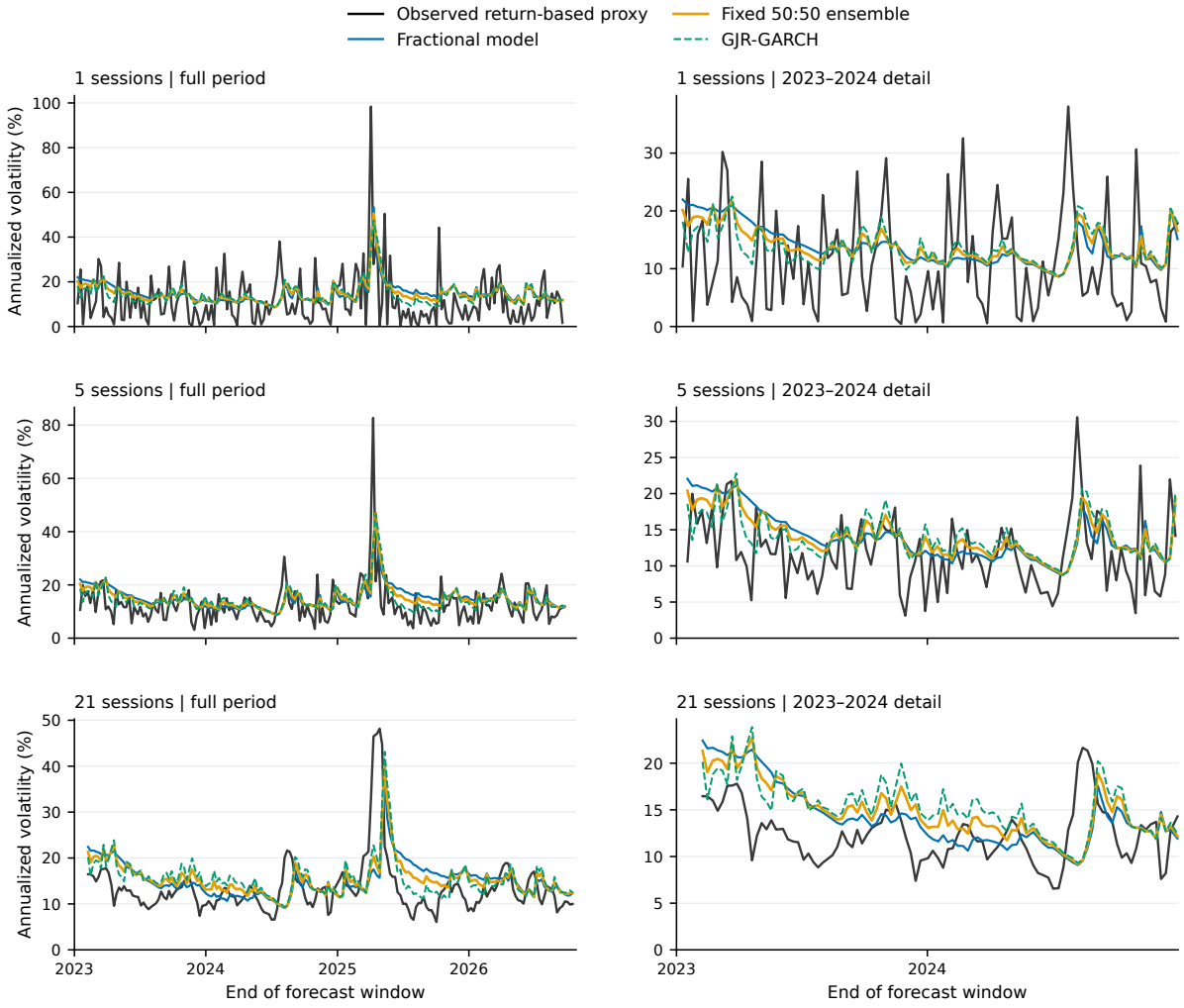


Figure 6: S&P 500 observed proxy and rolling predictions at 1, 5 and 21 sessions. Black denotes the subsequent return-based observation; blue is the persistent fractional model, orange the fixed ensemble, and dashed green GJR-GARCH. Each forecast was issued the stated number of observations before its plotted window-end date. Left panels retain the full period; right panels enlarge 2023–2024. Vertical ranges adapt to each panel, with all displayed peaks retained.

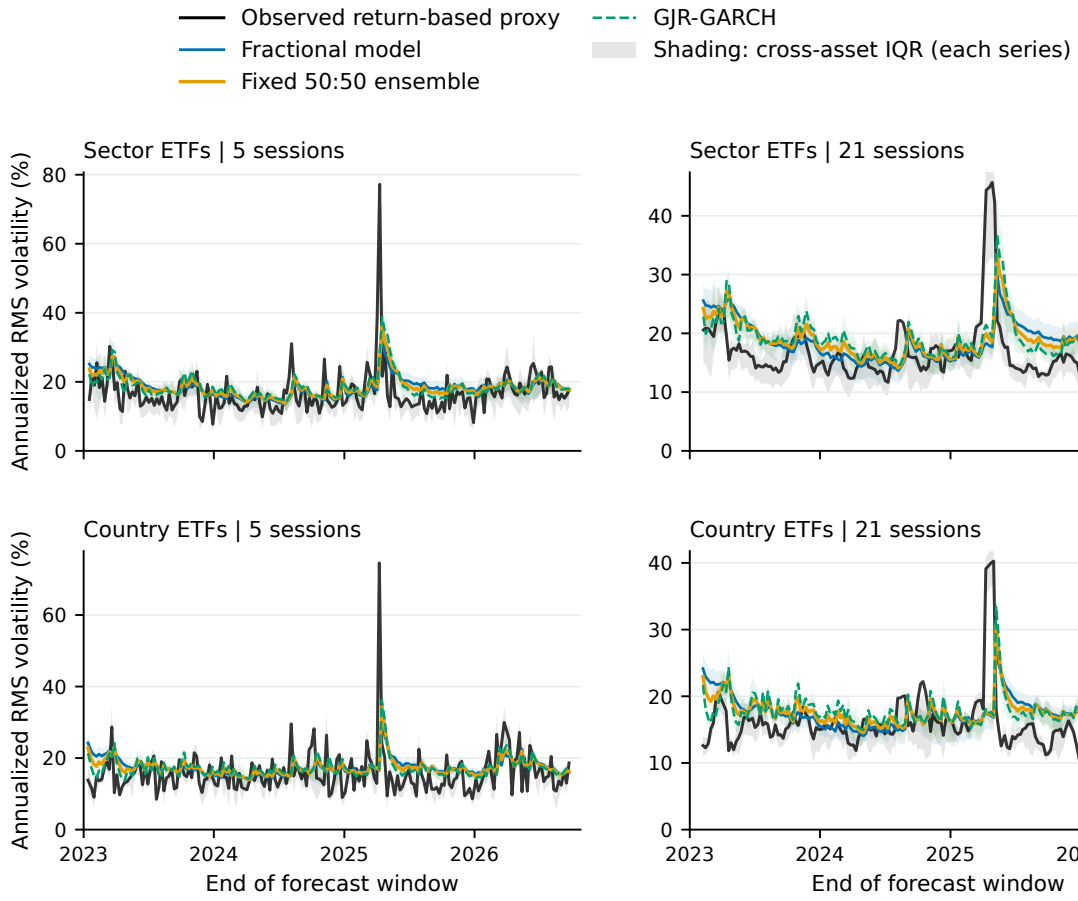


Figure 7: Observed and predicted risk evolution for all eleven sector ETFs and all eight country ETFs, at horizons of 5 and 21 sessions. Marginal variances are averaged across the complete panel before taking the square root. The lines show annualized root-mean-square marginal volatility. Shading in each series' color spans the cross-asset interquartile range of displayed volatilities at each target date; it describes dispersion across ETFs, not forecast uncertainty. Covariances are not included in this display.

Question	Recorded finding	Interpretation
Persistence, asymmetry and horizon selection	Lower mean loss in 6/6 US cases	Joint refinement; development data
Fixed ensemble vs. fractional / FIGARCH	Lower mean loss in 9/9 cases for each	Improvement relative to each reference
Fixed ensemble vs. GJR	Lower mean loss in 3/9 cases	All paired intervals include zero
Diffusion on country ETFs	No-graph variant better at all 3 horizons	No general diffusion benefit
Fractional vs. weighted Network GARCH	Lower loss in countries; higher in sectors	All 6 paired intervals include zero
Country one-session calibration	All 3 models reject the joint calibration null	Post-hoc HAC/Wald diagnostic

Table 5: Summary of the recorded findings. Lower mean losses describe this sample; inferential qualifications follow Section 6.1.

9 Interpretation, limitations and conclusions

9.1 What the restrictions explain

Theoretical contribution. The model restricts multivariate variance lag matrices to a common positive graph resolvent and a shared fractional profile. This structure yields an explicit impulse response and fixed-parameter moment bounds. It excludes asset-specific lag shapes and lag-dependent network geometry. The persistent state smooths the finite fractional profile geometrically, producing an infinite exponential tail when its coefficient is positive.

Empirical contribution. Adding persistence, asymmetry and multi-horizon selection reduces loss relative to the initial model in all six US comparisons. Diffusion and the fractional profile have less consistent effects across horizons and panels. The weighted Network GARCH benchmark has lower loss on sectors and higher loss on country ETFs than the standalone fractional model; all six paired intervals include zero at both block lengths.

Combining the refined model with GJR lowers pooled QLIKE relative to the fractional model and FIGARCH in all nine panel–horizon comparisons. Against GJR alone, the combination has lower loss in only three. All nine nominal 95% paired intervals contain zero with both five- and ten-origin blocks. These comparisons establish neither an advantage over GJR nor equivalent predictive performance. The DM tests also leave the GJR comparisons unresolved. Calibration regressions identify a one-session country-panel departure from the joint intercept-zero, slope-one null for all three examined models, including GJR (Appendix E). Table 5 collects the main findings.

9.2 Data vintages and prospective evaluation

At every origin, fitting uses only rows through that origin in the retained download. The download itself is retrospective: it does not recover the price version available at each historical forecast date.

To examine the effect of changing data vintages, let $P_t^{(T)}$ and $P_t^{(T')}$ be prices for date t from two download vintages, and set $a_t = P_t^{(T')}/P_t^{(T)} > 0$. Their log-returns satisfy

$$r_t^{(T')} - r_t^{(T)} = \log a_t - \log a_{t-1}. \tag{26}$$

A common multiplicative adjustment cancels within the affected span. Retrospective adjustment alone therefore does not imply look-ahead bias in every return. The relevant changes are those that alter the relative factor between adjacent dates, such as corrections to an event adjustment or to a historical close. The data interface documents missing dividend and split adjustments and price corrections [20]; this establishes possible mechanisms, not their occurrence in our panels.

A changed return can affect fitted means, scales, graphs and validation choices, as well as subsequent forecasts. Within a target window it also changes the realized proxy and loss. Measuring the direction and size of these effects requires multiple data vintages. We retain one vintage per panel, identified by its input hashes, so this sensitivity cannot be measured in the present study.

A prospective extension would freeze the model code, universe, refit rules, horizons, loss and observation schedule before collecting new targets. At each origin it would archive the data vintage, fitted state and timestamped forecasts before the first target session. A predefined rule would also fix when target returns are first recorded and how later corrections are reported. Missing releases, failed forecasts and any amendment would remain in the audit trail. Reporting would occur after a fixed collection period, regardless of the observed ranking. No such prospective run is included in the present results.

9.3 Scope and reproducibility

The positive variance floor and graph contraction constrain the recursion, but the forecast comparisons show that its shared structure does not help equally across panels and horizons. Daily squared-return targets are noisy, the conditional mean is constant within each fit, and the selected universes cover overlapping market shocks.

The implementation, tests and experiment records allow the results to be checked against the stated equations, given access to the retained inputs. Further evaluation would require richer volatility measurements and prospectively collected observations. Comparisons with network log-ARCH, threshold Network GARCH and dynamic-covariance models would test additional forms of temporal and cross-asset dependence.

A Implementation and reproduction

The initial implementation is in `fractional_diffusion.py`; the benchmark and asset generator are in `run_paper.py`. The follow-up uses `refined_fractional.py` and `run_`

`refinement.py`. The country-ETF extension uses `external_validation.py`, `download_external.py` and `run_external.py`.

The network comparison is implemented in `network_benchmark.py`. Run it with `run_network_benchmark.py`; its separate protocol is `paper/network_protocol.json`.

Existing return simulators and forecast heads remain separate. The recorded protocol is `paper/protocol.json`. Input hashes, candidate scores, graph estimates, selected parameters, optimizer statuses and each origin's forecasts are saved. Raw prices and detailed forecast records remain in ignored local artifacts.

```
python run_paper.py --synthetic --output artifacts/paper-synthetic
python run_paper.py --data-dir artifacts/yahoo-through-2026-10-16 \
  --output artifacts/paper-study --publish-assets
python run_refinement.py --synthetic --output artifacts/refinement-synthetic
python run_refinement.py --data-dir artifacts/yahoo-through-2026-10-16 \
  --output artifacts/refinement-study --publish-assets
python -m unittest discover -s tests -v
cd paper
tectonic --reruns 3 main.tex
```

The synthetic command uses 1,300 correlated Gaussian price observations with no imposed time-varying volatility or fractional mechanism. It is an offline execution example, not empirical evidence for the proposed model. The market snapshot download command and adjustment convention are documented in the data guide. External-study commands and source-package instructions are in `paper/README.md`.

The suite contains 66 tests on Python 3.12 under macOS. They check fractional weights, graph positivity and contraction, asset relabeling, GARCH/GJR filtering, Network GARCH gradients and the effective-memory impulse response. Conditional moments are compared with Monte Carlo simulation, and validation forecasts with fits on individually truncated histories. Perturbing future prices must leave earlier variances, parameter choices and graphs unchanged. Other tests cover FIGARCH library agreement, ensemble scoring, HAC and calibration calculations, multiplicity correction, paired-key integrity and degenerate inputs.

The presentation command `render_submission_assets.py` regenerates all eleven figures and the numerical tables from saved records, without refitting forecast models. It extends the path renderer `render_prediction_figures.py`. Input, source and output hashes are saved in `paper/submission_assets_manifest.json`. The additional diagnostics are reproduced with `run_forecast_diagnostics.py`; their protocol and computational amendment preserve the original analysis rules.

B Annual scores

C Relation to the original model

The original project used signed Grünwald–Letnikov coefficients in a return recurrence and a reflected nearest-neighbor sector Laplacian. Isolating the current value in a GL derivative does not yield that recurrence's lag and sign convention. A fractional derivative

	Index			Sectors		
Year	1	5	21	1	5	21
2023	0.0021	-0.0541	-0.0447	0.2066	0.1643	0.1675
2024	0.8771	0.2769	0.3731	0.3427	0.3209	0.3531
2025	1.2053	0.9798	1.3312	1.1208	0.9419	1.1219
2026	0.4703	0.5115	0.5196	0.9897	0.9254	0.9201

Table 6: Annual mean normalized QLIKE for the fractional graph model. Levels reflect both realized fluctuations and the origin's reference scale. Negative scores are permissible; comparisons should use common-origin loss differences.

also does not inherit the ordinary logarithmic chain rule. The present work therefore starts from (4)–(10), rather than invoking a fractional Black–Scholes equation.

The retained idea is memory plus diffusion. The changes are explicit: positive fractional-integral weights act on squared standardized innovations; a symmetric estimated graph replaces the artificial sector lattice; positive feedback controls second moments; and risk forecasts are compared chronologically against strong variance references. Hurst identification, convergence of a pricing PDE, and directional price-prediction superiority are not claimed.

D Observed and predicted paths for every ETF

The following figures show every ETF in the two recorded panels, using the 21-session horizon and the full evaluation period. Splitting the sector list across two pages only improves readability; assets are neither selected nor ranked by forecasting performance. The horizontal coordinate is the target window end, and the vertical coordinate is given by (25). Each asset has its own logarithmic vertical scale. All displayed values are strictly positive; no floor, peak clipping or extra temporal smoothing is applied.

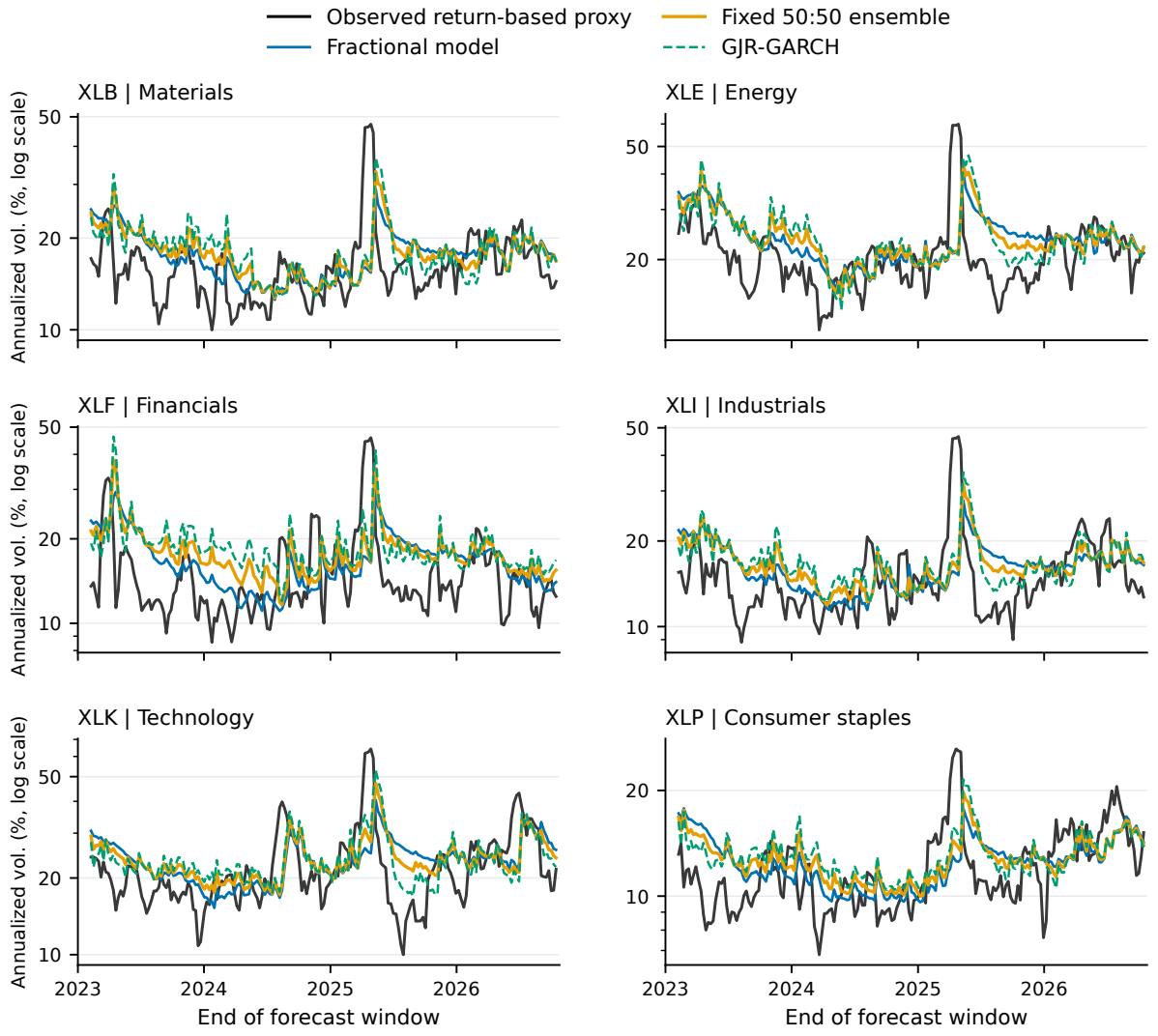


Figure 8: Twenty-one-session observed proxies and forecasts for materials, energy, financials, industrials, technology and consumer staples. Curves use the same methods and colors as Figure 6.

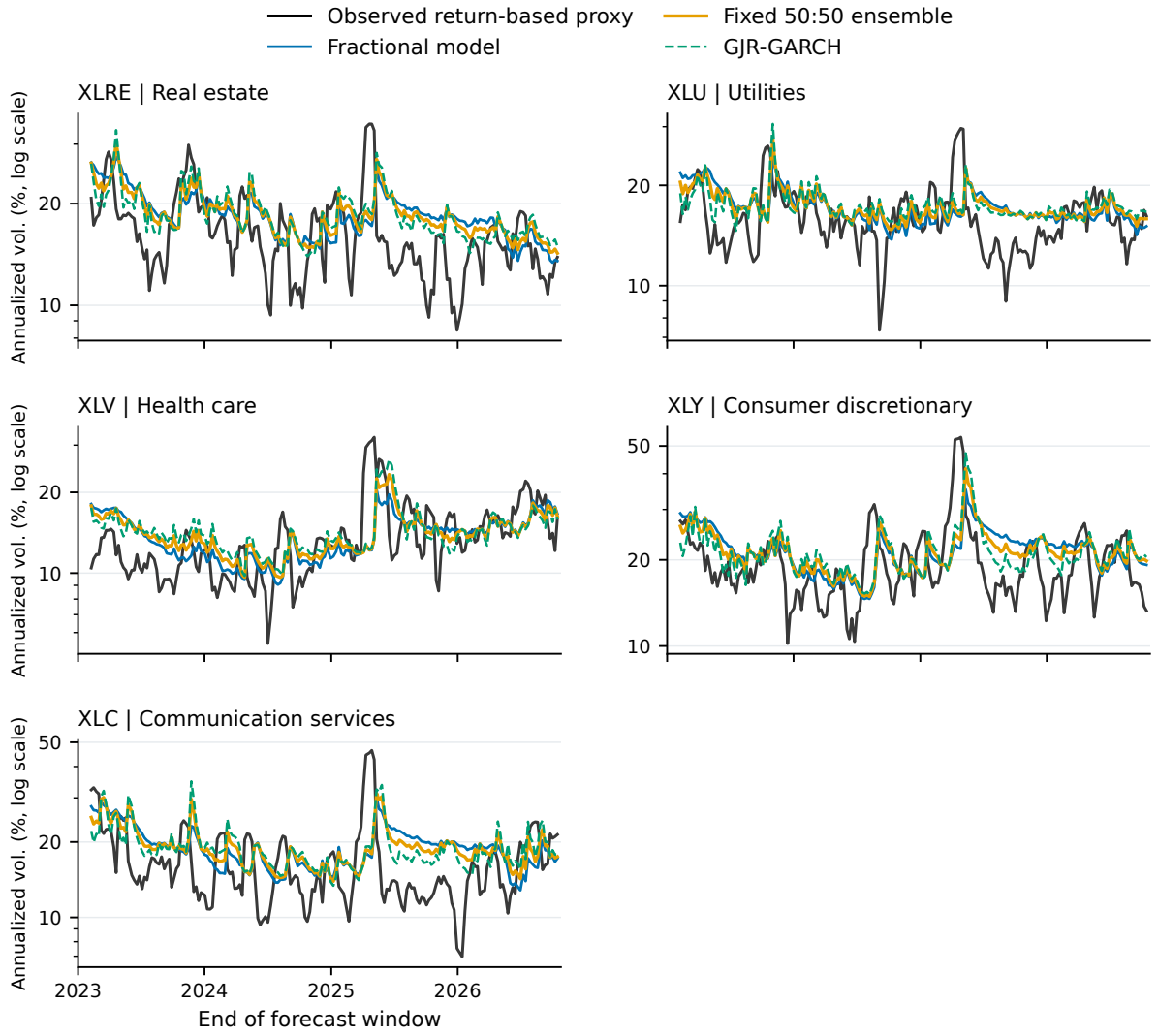


Figure 9: The remaining five sector ETFs: real estate, utilities, health care, consumer discretionary and communication services. Every sector is included exactly once across these two figures.

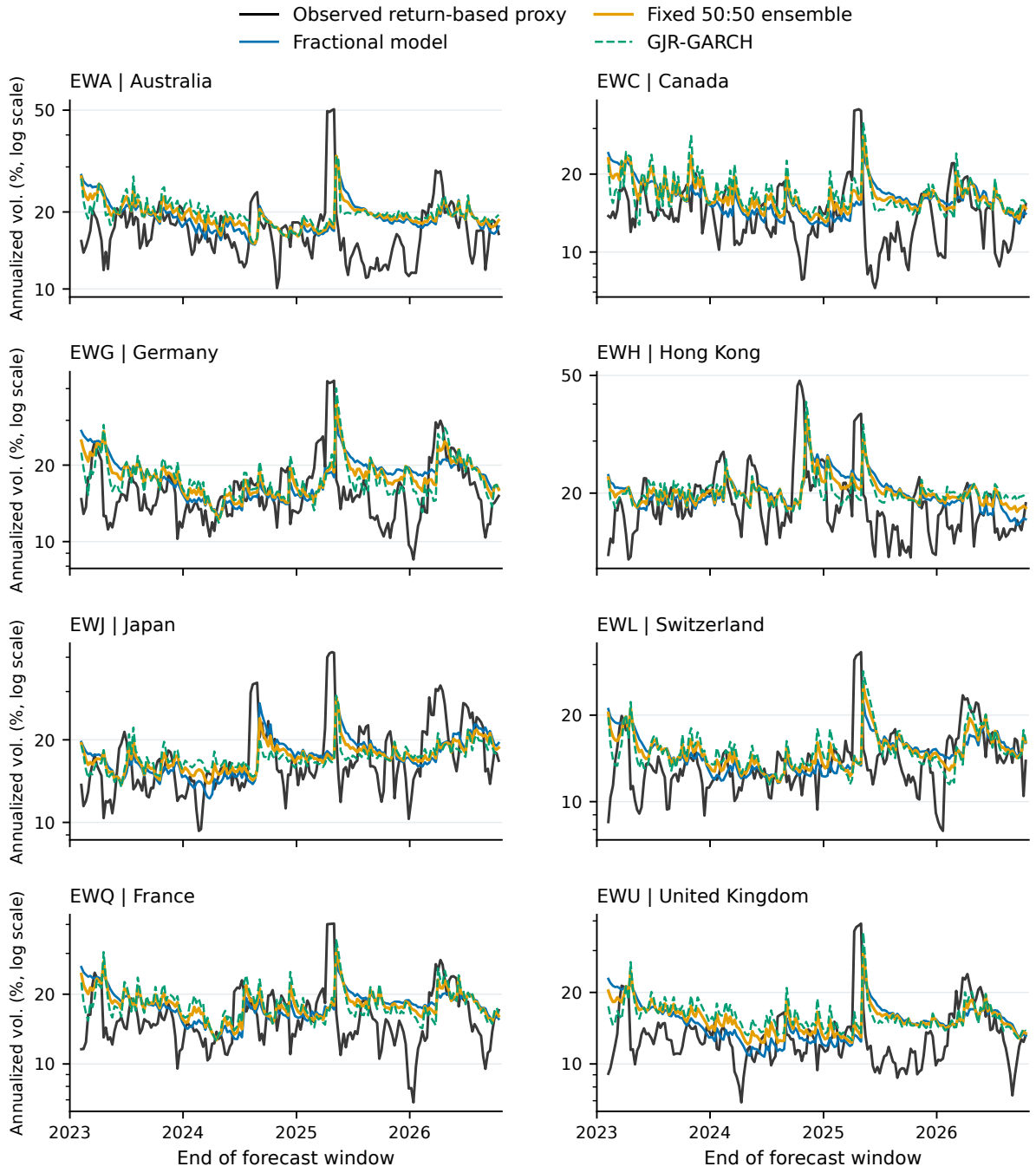


Figure 10: Twenty-one-session observed proxies and forecasts for all eight pre-specified country ETFs. The observed series is a noisy daily-return-based proxy; the plots do not imply prediction of the underlying ETF price direction.

E Additional forecast diagnostics

The diagnostics use the saved forecasts and are exploratory, since the earlier results were already known. Their protocol fixed the comparisons and inference rules before calculation. An amendment increased the tolerance for checking QLIKE reconstructed from decimal CSVs from 10^{-12} to 10^{-10}, before any diagnostic results were produced. The tests use the original stored losses, without refitting or recalibrating forecasts.

E.1 Paired predictive accuracy

For each panel and horizon, let d_o be the asset-average QLIKE difference between a candidate and GJR at origin o. Following Diebold and Mariano [6], we test a zero mean difference using

$$\mathrm{DM}_L = \frac{\bar{d}}{\sqrt{\widehat{\Omega}_L/n}}, \qquad \widehat{\Omega}_L = \widehat{\gamma}_0 + 2\sum_{\ell=1}^{L}\left(1-\frac{\ell}{L+1}\right)\widehat{\gamma}_\ell, \quad \widehat{\gamma}_\ell = \frac{1}{n}\sum_{o=\ell+1}^{n}(d_o-\bar{d})(d_{o-\ell}-\bar{d}).$$

The Bartlett HAC estimator follows Newey and West [13]. We use $n = 186$ origins and two-sided asymptotic normal p-values, with $L = 4$ as the primary choice and $L = 9$ as sensitivity. Lags count forecast origins, spaced five retained sessions apart: a 21-session target overlaps the next four origins. Asset losses are averaged before inference, preserving their contemporaneous dependence within each origin.

Table 7 reports both raw p-values and the Holm-adjusted [8] primary values across the 18 comparisons. Adjustment is also computed separately for the sensitivity analysis in the accompanying CSV. None of the 18 comparisons rejects equal expected loss at 5% with either lag choice, even before multiplicity adjustment. The asymptotic inference requires conditions on loss dependence and moments beyond the return second-moment proposition; researcher reuse remains as described in Section 6.1.

E.2 Calibration

For method m, average the normalized target and forecast over the J assets at each origin:

$$y_o = \frac{1}{J}\sum_{j=1}^{J}\frac{Y_{o,h,j}}{hs^2_{o,j}}, \qquad x_o^{(m)} = \frac{1}{J}\sum_{j=1}^{J}\frac{\widehat{V}^{(m)}_{o,h,j}}{hs^2_{o,j}}, \qquad y_o = \alpha + \beta x_o^{(m)} + e_o.$$

The Mincer–Zarnowitz regression [12, 17] diagnoses calibration under $H_0 : (\alpha, \beta) = (0, 1)$. Normalization uses the stored origin-specific reference scale. OLS coefficients are reported with Bartlett HAC standard errors, an $n/(n-2)$ covariance multiplier and a joint asymptotic χ^2_2 Wald test. The two lag choices are the same as for DM. Holm adjustment covers the 27 regressions separately at each lag choice.

The one-session country regressions reject the joint null for the fractional model, ensemble and GJR at both lag choices after adjustment. Other regressions do not reject at the adjusted 5% level. These are diagnostics of panel-average normalized second moments; aggregation can mask asset-level miscalibration. The fitted regression is not used to change any forecast.

Panel	h	Model	Δ	DM	p_4	p_9	Holm p_4
Index	1	Fractional	+0.05675	0.866	0.386	0.361	1.000
Index	1	Ensemble	+0.01419	0.329	0.742	0.729	1.000
Index	5	Fractional	+0.04454	1.837	0.066	0.074	1.000
Index	5	Ensemble	+0.01458	1.080	0.280	0.260	1.000
Index	21	Fractional	+0.06166	1.435	0.151	0.190	1.000
Index	21	Ensemble	+0.02200	1.174	0.240	0.297	1.000
Sectors	1	Fractional	+0.01889	1.425	0.154	0.144	1.000
Sectors	1	Ensemble	+0.00072	0.101	0.920	0.917	1.000
Sectors	5	Fractional	+0.00772	0.748	0.455	0.458	1.000
Sectors	5	Ensemble	-0.00293	-0.652	0.514	0.523	1.000
Sectors	21	Fractional	+0.01127	0.620	0.536	0.564	1.000
Sectors	21	Ensemble	-0.00050	-0.064	0.949	0.953	1.000
Countries	1	Fractional	+0.02065	1.392	0.164	0.130	1.000
Countries	1	Ensemble	+0.00474	0.622	0.534	0.487	1.000
Countries	5	Fractional	+0.01401	1.097	0.273	0.299	1.000
Countries	5	Ensemble	+0.00136	0.208	0.835	0.844	1.000
Countries	21	Fractional	+0.00921	0.826	0.409	0.454	1.000
Countries	21	Ensemble	-0.00092	-0.176	0.860	0.874	1.000

Table 7: Post-hoc predictive-accuracy diagnostics against GJR. Negative Δ favors the named model. DM uses $L = 4$; p_4 and p_9 use the indicated HAC lag lengths. Holm p_4 adjusts all 18 primary comparisons. Values are rounded only for display.

E.3 Win counts

Table 9 counts origins at which each asset's candidate forecast has lower QLIKE than GJR. Figure 11 answers a different question: how many assets have a lower full-period mean loss? Neither count is an independent-binomial sample. The CSV retains wins, ties and losses with a 10^{-12} tie tolerance.

Model	h	$\widehat{\alpha}$ (SE)	$\widehat{\beta}$ (SE)	R^2	p_4	p_9	Holm p_4
Index							
Fractional	1	-0.173 (0.334)	1.414 (0.549)	0.197	0.644	0.535	1.000
Ensemble	1	-0.212 (0.325)	1.499 (0.562)	0.204	0.551	0.414	1.000
GJR	1	-0.174 (0.298)	1.475 (0.552)	0.196	0.505	0.394	1.000
Fractional	5	-0.168 (0.270)	1.182 (0.474)	0.187	0.640	0.582	1.000
Ensemble	5	-0.150 (0.254)	1.165 (0.459)	0.202	0.610	0.548	1.000
GJR	5	-0.081 (0.220)	1.076 (0.423)	0.204	0.741	0.704	1.000
Fractional	21	0.174 (0.123)	0.709 (0.268)	0.088	0.353	0.242	1.000
Ensemble	21	0.125 (0.176)	0.761 (0.325)	0.136	0.750	0.595	1.000
GJR	21	0.156 (0.175)	0.702 (0.316)	0.164	0.627	0.416	1.000
Sectors							
Fractional	1	-0.590 (0.438)	1.937 (0.751)	0.092	0.354	0.188	1.000
Ensemble	1	-0.565 (0.444)	1.876 (0.739)	0.109	0.390	0.177	1.000
GJR	1	-0.441 (0.422)	1.684 (0.699)	0.115	0.533	0.286	1.000
Fractional	5	-0.166 (0.232)	1.210 (0.409)	0.104	0.435	0.351	1.000
Ensemble	5	-0.151 (0.247)	1.170 (0.412)	0.128	0.439	0.370	1.000
GJR	5	-0.066 (0.238)	1.039 (0.390)	0.138	0.565	0.558	1.000
Fractional	21	0.164 (0.104)	0.756 (0.200)	0.085	0.288	0.369	1.000
Ensemble	21	0.100 (0.159)	0.818 (0.267)	0.124	0.794	0.738	1.000
GJR	21	0.137 (0.184)	0.747 (0.289)	0.142	0.668	0.554	1.000
Countries							
Fractional	1	0.314 (0.090)	0.441 (0.112)	0.041	< 0.001	< 0.001	< 0.001
Ensemble	1	0.277 (0.093)	0.489 (0.115)	0.059	< 0.001	< 0.001	< 0.001
GJR	1	0.280 (0.101)	0.485 (0.126)	0.071	< 0.001	< 0.001	< 0.001
Fractional	5	0.291 (0.161)	0.607 (0.167)	0.020	0.061	0.085	1.000
Ensemble	5	0.296 (0.178)	0.600 (0.177)	0.024	0.062	0.061	1.000
GJR	5	0.340 (0.171)	0.542 (0.162)	0.025	0.012	0.008	0.287
Fractional	21	0.344 (0.169)	0.527 (0.228)	0.038	0.105	0.135	1.000
Ensemble	21	0.323 (0.213)	0.545 (0.263)	0.046	0.219	0.164	1.000
GJR	21	0.366 (0.218)	0.482 (0.255)	0.047	0.111	0.050	1.000

Table 8: Calibration of normalized panel-average forecasts. Parentheses contain HAC standard errors with $L = 4$. The joint null is $(\alpha, \beta) = (0, 1)$; p_4 and p_9 are raw Wald values, and Holm p_4 adjusts all 27 primary regressions. Small p-values indicate a departure from calibration, not a better forecasting model.

	Fractional wins			Ensemble wins		
Asset	1	5	21	1	5	21
Index						
SP500	79	84	92	80	86	98
Sectors						
XLB	83	108	111	86	110	114
XLE	85	89	99	89	93	102
XLF	91	100	109	95	109	116
XLI	87	97	101	91	100	106
XLK	84	86	95	90	90	103
XLP	106	100	98	107	106	102
XLRE	96	85	82	96	89	87
XLU	97	102	91	98	104	98
XLV	97	111	109	100	118	118
XLY	89	98	86	91	103	88
XLC	80	86	80	84	93	83
Countries						
EWA	85	91	100	86	96	101
EWC	89	85	96	96	95	101
EWG	73	85	85	73	90	89
EWH	98	95	89	99	97	94
EWJ	83	92	94	85	97	102
EWL	88	89	99	90	94	107
EWQ	85	89	99	88	91	107
EWU	87	79	95	90	83	98

Table 9: Per-asset counts of lower-loss origins against GJR, out of 186 paired origins at each horizon. Fractional denotes the refined model. Counts describe frequency; they do not measure the size of gains or losses.

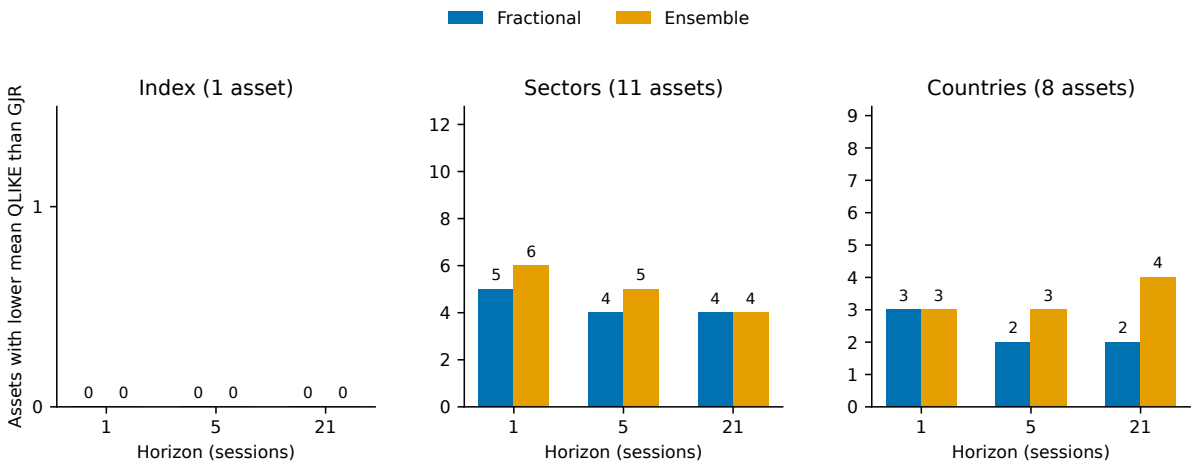


Figure 11: Number of assets with lower full-period mean QLIKE than GJR, by panel and horizon. Denominators are one index, eleven sectors and eight country ETFs. The complete universe is retained in each comparison.

Data and code availability

Code, tests, protocols and aggregate results are available at https://github.com/enia-ardid/mathematical_finance. The code and manuscript should be identified by their Git commit. Raw prices and detailed origin-level records are retained locally with source metadata and hashes and are not bundled. Exact numerical reproduction requires those snapshots; a fresh adjusted-price download may differ.

Declaration of competing interests

We declare no competing financial interests or personal relationships that could have influenced this work.

References

[1] Richard T. Baillie, Tim Bollerslev, and Hans Ole Mikkelsen. Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 74(1):3–30, 1996. doi: 10.1016/S0304-4076(95)01749-6.

[2] Ole E. Barndorff-Nielsen and Neil Shephard. Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 64(2):253–280, 2002. doi: 10.1111/1467-9868.00336.

[3] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986. doi: 10.1016/0304-4076(86)90063-1.

[4] Tim Bollerslev, Andrew J. Patton, and Rogier Quaedvlieg. Exploiting the errors: A simple approach for improved volatility forecasting. *Journal of Econometrics*, 192 (1):1–18, 2016. doi: 10.1016/j.jeconom.2015.10.007.

[5] Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196, 2009. doi: 10.1093/jjfinec/nbp001.

[6] Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995. doi: 10.1080/07350015.1995.10524599.

[7] Lawrence R. Glosten, Ravi Jagannathan, and David E. Runkle. On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5):1779–1801, 1993. doi: 10.1111/j.1540-6261.1993.tb05128.x.

[8] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. doi: 10.2307/4615733.

[9] J. R. M. Hosking. Fractional differencing. *Biometrika*, 68(1):165–176, 1981. doi: 10.1093/biomet/68.1.165.

[10] Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004. doi: 10.1016/S0047-259X(03)00096-4.

[11] Raffaele Mattera and Philipp Otto. Network log-ARCH models for forecasting stock market volatility. *International Journal of Forecasting*, 40(4):1539–1555, 2024. doi: 10.1016/j.ijforecast.2024.01.002.

[12] Jacob Mincer and Victor Zarnowitz. The evaluation of economic forecasts. In Jacob Mincer, editor, *Economic Forecasts and Expectations: Analysis of Forecasting Behavior and Performance*, pages 3–46. National Bureau of Economic Research, 1969. URL https://www.nber.org/system/files/chapters/c1214/c1214.pdf.

[13] Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55 (3):703–708, 1987. doi: 10.2307/1913610.

[14] Philipp Otto, Wolfgang Schmid, and Robert Garthoff. Generalised spatial and spatiotemporal autoregressive conditional heteroscedasticity. *Spatial Statistics*, 26:125–145, 2018. doi: 10.1016/j.spasta.2018.07.005.

[15] Yue Pan and Jiazhu Pan. Threshold network GARCH model. *Journal of Time Series Analysis*, 45(6):910–930, 2024. doi: 10.1111/jtsa.12743.

[16] Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256, 2011. doi: 10.1016/j.jeconom.2010.03.034.

[17] Andrew J. Patton and Kevin Sheppard. Evaluating volatility and correlation forecasts. In Torben Gustav Andersen, Richard A. Davis, Jens-Peter Kreiß, and Thomas Mikosch, editors, *Handbook of Financial Time Series*, pages 801–838. Springer, 2009. doi: 10.1007/978-3-540-71297-8_36.

[18] David I. Shuman, Sunil K. Narang, Pascal Frossard, Antonio Ortega, and Pierre Vandergheynst. The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains. *IEEE Signal Processing Magazine*, 30(3):83–98, 2013. doi: 10.1109/MSP.2012.2235192.

[19] Allan Timmermann. Forecast combinations. In Graham Elliott, Clive W. J. Granger, and Allan Timmermann, editors, *Handbook of Economic Forecasting*, volume 1, pages 135–196. Elsevier, 2006. doi: 10.1016/S1574-0706(05)01004-9.

[20] yfinance contributors. Price repair. yfinance documentation, n.d. URL https://ranaroussi.github.io/yfinance/advanced/price_repair.html. Accessed 23 October 2026.

[21] Jing Zhou, Dong Li, Rui Pan, and Hansheng Wang. Network GARCH model. *Statistica Sinica*, 30(4):1723–1740, 2020. doi: 10.5705/ss.202018.0234. URL https://www3.stat.sinica.edu.tw/statistica/j30n4/j30n403/j30n403.html.

[22] Peiyi Zhou. A new implementation of network GARCH model for stock volatility and co-volatility forecasting. *Journal of Forecasting*, 2026. doi: 10.1002/for.70191. Published online 12 July 2026.